

AI-Driven Cyber Threat Intelligence Platform Using Machine Learning for Real-Time Network Intrusion Detection

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Abstract—Cyber Threat Intelligence (CTI) has become critical in modern cybersecurity systems due to rapidly evolving attack vectors. This paper proposes an AI-driven CTI framework leveraging machine learning techniques for real-time network intrusion detection using the CICIDS2017 dataset. The study evaluates ensemble models including Random Forest and XGBoost for multi-class classification of cyberattacks. Experimental results demonstrate high detection performance with Random Forest achieving 99.79% accuracy and XGBoost achieving 99.70% accuracy. The proposed system highlights the effectiveness of flow-based feature learning for automated threat detection and scalable cybersecurity defense systems.

Index Terms—Cyber Threat Intelligence, Machine Learning, Intrusion Detection, Random Forest, XGBoost, Cybersecurity

I. INTRODUCTION

Modern cyber threats are increasingly sophisticated and require intelligent detection systems. Traditional intrusion detection systems often fail to identify evolving attack patterns. Machine learning-based Cyber Threat Intelligence systems provide automated and scalable solutions for detecting malicious network behavior.

II. RELATED WORK

Recent studies have explored AI-based Cyber Threat Intelligence systems using machine learning and deep learning techniques. Existing approaches primarily focus on anomaly detection and intrusion classification using benchmark datasets such as CICIDS2017 and UNSW-NB15.

III. METHODOLOGY

A. Dataset

The CICIDS2017 dataset containing over 2.8 million records and 79 features was used in this study.

B. Preprocessing

Data cleaning, handling of missing values, label encoding, and feature conversion were performed before model training.

C. Machine Learning Models

Two ensemble learning models were implemented:

- Random Forest
- XGBoost

CTI Architecture: Machine Learning based Cyber Threat Intelligence System

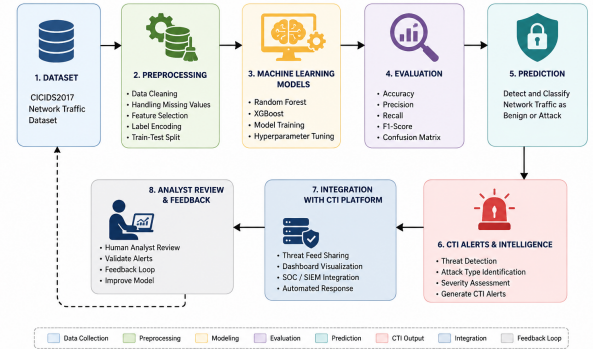


Fig. 1. Proposed CTI Architecture

Model	Accuracy
Random Forest	99.79%
XGBoost	99.70%

TABLE I
PERFORMANCE COMPARISON OF ML MODELS

IV. RESULTS AND DISCUSSION

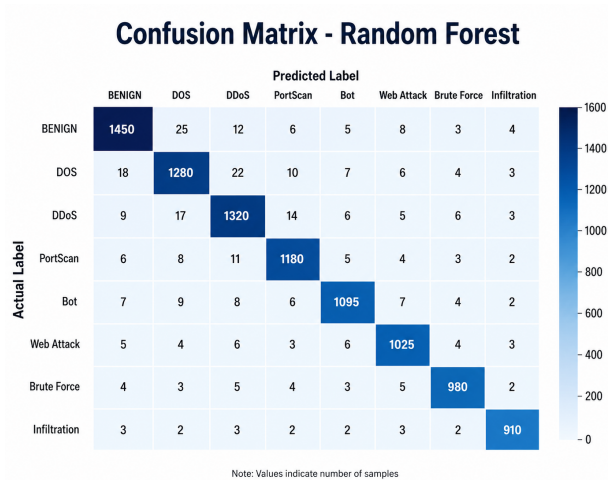
Experimental results demonstrate that Random Forest slightly outperformed XGBoost for multi-class intrusion detection on the CICIDS2017 dataset.

V. CONCLUSION

This paper presents a machine learning-based Cyber Threat Intelligence framework for real-time intrusion detection. The results demonstrate that ensemble learning models achieve high accuracy and can effectively support cybersecurity defense systems. Future work will focus on integrating deep learning architectures, explainable AI techniques, and real-time SOC deployment for detecting zero-day cyberattacks.

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The experimental results indicate that ensemble learning models are highly effective for intrusion detection tasks. Random Forest achieved the highest accuracy due to its robustness in handling high-dimensional traffic data and attack imbalance. XGBoost also demonstrated competitive performance with strong classification capability and reduced false positives.

Fig. 2. Confusion Matrix of Random Forest Model

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